

# Paipu Genome Preparation

## Pipeline Set Up

*Set up these elements prior to running the pipeline.*

- 1. Download DEXSeq script
- 2. Replace paths/slurm information in the Nextflow pipeline (*run\_genome\_prep\_slurm.sh*)
- 3. Create Input File EX:

Ailuropoda melanoleuca  
Antilocapra americana  
Aotus nancymae

## Query Genome Information

Query reference genome information for each mammal in input file.

ncbi\_queries/\${mammal}.json

## Parse Genome Information

Parse queried genome information to find the most recent genome for each mammal and available gene annotation files.

output\_all.tsv

output.tsv

## Download Mammal Reference Genome

Download reference genome and gene annotation files identified by genome query for each mammal

genomic.fna

genomic.gtf

## Prep Genome Indexes

Prep FAI index, sequence dictionary, and HISAT index

## Prepare GFF File

dexseq\_prepare\_annotation.py

Prepare GFF files for DEXSeq

Genomes are ready for Paipu!